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MAC286 Assignment

(from book) Write a method for the Queue class that displays the contents of the queue. Note that this does not mean simply displaying the contents of the underlying array. You should show the queue contents from the first item inserted to the last, without indicating to the viewer whether the sequence is broken by wrapping around the end of the array. Be careful that one item and no items display properly, no matter where **front** and **rear** are.

NOTE: Display both the Array content and the queue content.

The partial Queue class is given below. You need to add the **display()** method to this class.

Also the DisplayApp class is given below. You will use this class to test your answer.

A sample output when this program runs is:

```
Enter first letter of:
show, insert, remove, exit: i
Enter value to insert: 5
Array: 5 0 0 0 0
Queue: 5

Enter first letter of:
show, insert, remove, exit: i
Enter value to insert: 3
Array: 5 3 0 0 0
Queue: 5 3

Enter first letter of:
show, insert, remove, exit: r
Removed 5
Array: 5 3 0 0 0
Queue: 3

Enter first letter of:
show, insert, remove, exit: e
```

Note: the program should work for any integer arrays of small size.

What to submit on Brightspace:

- A single text document will the code for the display() method ONLY.
- A short screen recording demonstrating and explaining your code.

```java
class Queue
{
	private final int maxSize;
	private long[] queArray;
	private int front;
	private int rear;
	private int nItems;
	//-------------------------------------------------------------
	public Queue(int s)          // constructor
	{
		maxSize = s;
		queArray = new long[maxSize];
		front = 0;
		rear = -1;
		nItems = 0;
	}
	//-------------------------------------------------------------
	public void insert(long j)   // put item at rear of queue
	{
		if(rear == maxSize-1)         // deal with wraparound
			rear = -1;
		queArray[++rear] = j;         // increment rear and insert
		nItems++;                     // one more item
	}
	//-------------------------------------------------------------
	public long remove()         // take item from front of queue
	{
		long temp = queArray[front++]; // get value and incr front
		if(front == maxSize)           // deal with wraparound
			front = 0;
		nItems--;                      // one less item
		return temp;
	}
	//-------------------------------------------------------------
	public long peekFront()      // peek at front of queue
	{ return queArray[front]; }
	//-------------------------------------------------------------
	public boolean isEmpty()     // true if queue is empty
	{ return (nItems==0); }
	//-------------------------------------------------------------
	public boolean isFull()      // true if queue is full
	{ return (nItems==maxSize); }
	//-------------------------------------------------------------

}  // end class Queue
```

```java
import java.io.*;

class DisplayApp
{
	public static void main(String[] args) throws IOException
	{
		Queue theQueue = new Queue(5);  // queue holds 5 items
		long value;

		while(true)
		{
			System.out.println("Enter first letter of:");
			System.out.print("show, insert, remove, exit: ");
			int choice = getChar();
			switch(choice)
			{
			case 's':
				theQueue.display();
				break;
			case 'i':
				if( !theQueue.isFull() )
				{
					System.out.print("Enter value to insert: ");
					value = getLong();
					theQueue.insert(value);
					theQueue.display();
				}
				else
					System.out.print("*** Queue is full ***\n");
				break;
			case 'r':
				if( !theQueue.isEmpty() )
				{
					value = theQueue.remove();
					System.out.println("Removed " + value);
					theQueue.display();
				}
				else
					System.out.print("*** Queue is empty ***\n");
				break;
			case 'e':
				System.exit(0);
			default:
				System.out.print("Invalid entry\n");
			}  // end switch
		}  // end while
	}  // end main()
	//-------------------------------------------------------------
	public static char getChar() throws IOException
	{
		String s = getString();
		if(s.length() == 0)     // if immediate Enter
			return '#';
		else
			return s.charAt(0);
	}
	//-------------------------------------------------------------
	public static long getLong() throws IOException
	{
		String s = getString();
		return (long)Integer.parseInt(s);
	}
	//-------------------------------------------------------------
}  // end class DisplayApp
```